

Interpret Division of Fractions

Lesson 2-1

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

In $3 \div 1/4 = 12$, the dividend is 3 — it is the total being split

Dividend

The number you are splitting up in a division problem.

In $3 \div 1/4 = 12$, the divisor is $1/4$ — it is the size of each piece

Divisor

The number you split by in a division problem.

In $3 \div 1/4 = 12$, the quotient is 12 — there are 12 quarter-size pieces in 3

Quotient

The answer when you divide.

The reciprocal of $1/4$ is $4/1 = 4$. Multiplying by the reciprocal gives the same result as dividing.

Reciprocal

A fraction turned upside down.

$1/2$, $1/3$, $1/4$, $1/5$ — each represents one equal part of a whole

Unit fraction

A fraction with 1 on top, like $1/4$.

Key Ideas & Notes

- Agent Reyes finds a 3-foot strip of tape at the crime scene.
- The lab needs the tape cut into $\frac{1}{4}$ -foot sections for analysis.
- How many sections can be made?
- The answer cracks the code to the locked evidence vault.
- Use the bar model to see how many $\frac{1}{4}$ -size pieces fit into 3 wholes.

Think About It

- What total length is being divided?
- What is the size of each section?
- Will the answer be greater than or less than 3?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Agent Reyes cuts a 3-foot strip of tape into $\frac{1}{4}$ -foot sections. Before calculating, will the number of sections be more or less than 3? Why?

Level 1 support

Start here: There will be more than 3 sections because each piece is smaller than a whole foot.

The number of sections will be ____ than 3 because ____.

El número de secciones será ____ que 3 porque ____.

Each $\frac{1}{4}$ -foot piece is ____ than one whole foot, so ____.

Cada pieza de $\frac{1}{4}$ de pie es ____ que un pie entero, así que ____.

Word bank: **dividend** **divisor** **unit fraction**

Listen for: Students reason that dividing by a number less than 1 gives a quotient greater than the dividend, so there are more than 3 sections.

Level 2

Predict how the number of sections would change if each piece were $\frac{1}{8}$ foot instead of $\frac{1}{4}$ foot. Justify your prediction.

- With $\frac{1}{8}$ -foot pieces there would be ____ sections because ____.
- A smaller divisor produces ____ because ____.

EXPLORE

What does $3 \div \frac{1}{4}$ actually mean? Show your partner with the bar model.

Level 1 support

Start here: $3 \div \frac{1}{4}$ asks how many $\frac{1}{4}$ -size pieces fit into 3.

Dividing 3 by $\frac{1}{4}$ means finding how many ____ fit into ____.

Dividir 3 entre $\frac{1}{4}$ significa encontrar cuántos ____ caben en ____.

My bar model shows ____ fourths, so the quotient is ____.

Mi modelo de barras muestra ____ cuartos, así que el cociente es ____.

Word bank: **dividend** **divisor** **quotient** **unit fraction**

Listen for: Students name 3 as the dividend and $\frac{1}{4}$ as the divisor, and explain the model shows 12 fourths fit into 3.

Level 2

Why is $3 \div \frac{1}{4}$ NOT the same as $\frac{1}{4}$ of 3? Critique a classmate who confuses the two.

- $3 \div \frac{1}{4}$ is different from $\frac{1}{4}$ of 3 because ____.
- Taking $\frac{1}{4}$ of 3 gives ____, but dividing gives ____.

CONNECT

Where in cooking, sharing, or building might you divide a whole by a fraction?

Level 1 support

Start here: Any time you ask 'how many small pieces fit' you are dividing by a fraction.

In real life I divide by a fraction when ____.

En la vida real divido entre una fracción cuando ____.

For example, ____ uses ____ $\div$ ____.

Por ejemplo, ____ usa ____ $\div$ ____.

Word bank: **dividend** **divisor** **quotient**

Listen for: Students give a 'how many fit' scenario and correctly identify the whole as the dividend and the piece size as the divisor.

Level 2

Generalize: how can you tell from the words of a problem that you should divide by a fraction rather than multiply?

- The clue that tells me to divide is ____.
- If the problem instead asked for ____, I would multiply because ____.

Guided Examples

Example 1

What does $3 \div 1/4$ mean?

Solution: $3 \div 1/4$ asks: how many $1/4$ -size pieces fit into 3? The answer is 12, because each whole has 4 fourths, and $3 \times 4 = 12$.

Answer: A. How many $1/4$ -size pieces fit into 3

Example 2

Which expression means 'how many $1/3$ -size pieces are in 2'?

Solution: Finding how many $1/3$ -size pieces fit into 2 is represented by $2 \div 1/3$. The answer is 6.

Answer: A. $2 \div 1/3$

Example 3

A rope is 4 feet long. How many $1/2$ -foot pieces can be cut from it?

Solution: $4 \div 1/2 = 4 \times 2 = 8$ pieces. Each whole foot contains 2 halves, and $4 \times 2 = 8$.

Answer: A. 8 pieces

Write About the Math

The Writing Revolution

I can explain a fraction division using the words dividend, divisor, quotient, reciprocal, and unit fraction.

1. Kernel Sentence subject + verb

Model: Dividend is the number you are splitting up in a division problem.

Dividendo es el número que estás repartiendo en una división.

Write a kernel sentence about dividend. Use a subject and a verb.

Escribe una oración base sobre dividendo. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Dividend matters in math

Dividendo importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Dividend matters in math because ____.

Dividendo importa en matemáticas porque ____.

but
pero

Dividend matters in math, but ____.

Dividendo importa en matemáticas, pero ____.

so
entonces

Dividend matters in math, so ____.

Dividendo importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about dividend.

Di un hecho verdadero sobre dividend.

Dividend ____.

Question *Pregunta*

Ask a question about dividend.

Haz una pregunta sobre dividend.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about dividend.

Muestra entusiasmo sobre dividend.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with dividend.

Dile a un compañero qué hacer con dividend.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

The number of sections will be ____ **than 3 because** ____.

El número de secciones será ____ *que 3 porque* ____.

Each 1/4-foot piece is ____ **than one whole foot, so** ____.

Cada pieza de 1/4 de pie es ____ *que un pie entero, así que* ____.

Dividing 3 by 1/4 means finding how many ____ **fit into** ____.

Dividir 3 entre 1/4 significa encontrar cuántos ____ *caben en* ____.

Try It

Solve on your own. Check the answer key when you are done.

1. What is $6 \div 1/5$?

- A. 30
- B. $6/5$
- C. $5/6$
- D. 1

Show your work:

2. A shelf is 5 feet long. Books are each $1/4$ foot wide. How many books fit on the shelf? Which expression represents this?

- A. $5 \div 1/4 = 20$ books
- B. $5 \times 1/4 = 1 \frac{1}{4}$ books
- C. $1/4 \div 5 = 1/20$ book
- D. $5 + 1/4 = 5 \frac{1}{4}$ books

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Dividing 6 by $\frac{1}{2}$ gives 12, but dividing 6 by 2 gives 3. How can dividing by a smaller number ($\frac{1}{2}$) give a bigger answer than dividing by a larger number (2)? Explain using a real-world example.

Sentence starter: When I divide 6 by 2, I am _____. When I divide 6 by $\frac{1}{2}$, I am _____. The answer is bigger because _____.

Show your work:

Reflect — Exit Ticket

What does $5 \div \frac{1}{4}$ mean, and what is the quotient?

- A. How many $\frac{1}{4}$ -size pieces fit into 5; quotient is 20
- B. 5 groups of $\frac{1}{4}$; quotient is $\frac{5}{4}$
- C. $\frac{1}{4}$ of 5; quotient is $\frac{5}{4}$
- D. 5 minus $\frac{1}{4}$; quotient is $4\frac{3}{4}$

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. 30 — $6 \div 1/5$ means how many $1/5$ -size pieces fit into 6. Each whole has 5 fifths, so $6 \times 5 = 30$.
2. **Try It 2:** A. $5 \div 1/4 = 20$ books — You are finding how many $1/4$ -foot pieces fit into 5 feet, which is $5 \div 1/4 = 5 \times 4 = 20$ books.
3. **Exit Ticket:** A. How many $1/4$ -size pieces fit into 5; quotient is 20 — $5 \div 1/4$ means 'how many $1/4$ -size pieces fit into 5.' Since each whole has 4 fourths, $5 \times 4 = 20$.

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Dividend is the number you are splitting up in a division problem.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).